



TECHNICAL CLARIFICATION REQUEST

AshGridX API Integration and Pilot Readiness Review

Device control, tamper detection, webhooks, security, device onboarding, and operational assurance

Prepared for	AshGridX Technical Team
Prepared by	BuiltRight Services Ltd
Date	11 August 2026
Document status	Response requested for pilot integration

Purpose: confirm the missing technical and operational details required to integrate AshGridX safely into BuiltRight's financed solar installation and asset-control platform.

Reference reviewed

Ashgrid API Documentation - Integration guide, endpoint specifications, and webhook reference (9 pages, supplied 11 August 2026).

1 Executive summary

BuiltRight intends to use AshGridX devices in financed solar installations. Each physical device will be assigned to a specific customer, project, site, and financed asset. BuiltRight requires secure remote activation and disablement, auditable control history, immediate tamper alerts, and reliable identification of the affected customer device.

The supplied documentation is a useful starting point. It confirms API-key authentication, staging and production environments, device on/off control, control history, protection removal, protection history, and signed webhooks. However, several details required for safe pilot deployment and later production automation are not yet documented.

Priority clarification: the example webhook event type BYPASS is not defined. The document does not confirm that unplugging a cable, disconnecting the inverter, opening the device, or otherwise tampering with the installation will trigger an immediate alert.

Requested outcome

- Written answers to the questions in this document.
- Updated API documentation or an OpenAPI/Postman collection covering the missing operations and event definitions.
- Staging credentials and at least one pilot or simulated device.
- Confirmation of the expected pilot test procedure and support contact.

2 Capabilities confirmed by the supplied documentation

Area	What the document states	Review status
Environment	Staging and production base URLs are provided.	Confirmed
Authentication	API key supplied in the Authorization request header.	Confirmed
Device control	POST /device/control with customerDeviceId and control: on off.	Confirmed
Control history	POST /device/control/history with pagination and device/site filters.	Confirmed
Protection removal	POST /device/remove-protection using customerDeviceId.	Confirmed, meaning unclear
Protection history	GET /device/protection-history using query parameters.	Confirmed
Webhooks	Asynchronous POST requests with an HMAC SHA-256 signature header.	Confirmed, incomplete specification
Webhook example	Payload contains type, deviceId, and time. Example type is BYPASS.	Confirmed, event undefined

Documented device-control request

```
POST /device/control
Authorization: <YOUR_API_KEY>
Content-Type: application/json
```

```
{
  "customerDeviceId": "string",
  "control": "on | off"
}
```

Response: 204 No Content

Observation: a 204 response proves that the API accepted the request, but the documentation does not state whether it proves that the physical device or inverter actually changed state.

3 Device onboarding, assignment, and identity

BuiltRight must be able to add more devices as financed installations increase. The current document contains device information in history responses, but it does not document how devices are registered, assigned, listed, replaced, or transferred.

1. How does BuiltRight register a new device through the API?
2. Can device registration and customer assignment be automated, or must they be performed in the AshGridX dashboard?
3. Is customerDeviceId generated by AshGridX or supplied by BuiltRight?
4. What is the relationship between deviceId, customerDeviceId, deviceNumber, inventoryNumber, siteId, ownerId, and accountNumber?
5. The control endpoint uses customerDeviceId, while the webhook example uses deviceId. How should BuiltRight map these values reliably?
6. Is there an endpoint to list all devices belonging to BuiltRight and retrieve their assigned customers/sites?
7. Is there an endpoint to retrieve one device by customerDeviceId, deviceId, or deviceNumber?
8. How is a device replaced while preserving the customer, project, and control-history relationship?
9. Can one customer or installation have multiple AshGridX devices?
10. Can a device be unassigned, decommissioned, or reassigned through the API?
11. What fields are mandatory when a customer device is created, and which customer information is stored by AshGridX?

Minimum BuiltRight mapping requirement

Mapping group	Required fields
AshGridX identity	deviceId, customerDeviceId, deviceNumber, siteId
BuiltRight identity	customer ID, financing request, order, installation, site/address
Operational state	requested state, confirmed physical state, connectivity, last-seen time
Lifecycle	assigned, installed, active, disabled, replaced, decommissioned

4 Remote control, status verification, and protection

1. Does a 204 response from /device/control mean the command was queued, transmitted, acknowledged by the device, or physically executed?
2. How can BuiltRight retrieve the device's live on/off state after issuing a command?
3. Does device-control history represent requested state or confirmed physical execution?
4. Is there a command reference or transaction ID that can be used for idempotency, reconciliation, and support?
5. What happens when a device is offline when an on/off command is submitted?
6. Are commands queued for later execution, and if so, for how long?
7. Can control commands fail after the API initially returns success? How is that failure reported?
8. Is there a webhook for CONTROL_ACCEPTED, CONTROL_EXECUTED, CONTROL_FAILED, or CONTROL_TIMED_OUT?
9. Are repeated requests idempotent, and how should BuiltRight prevent duplicate commands?

10. Is bulk device control supported or planned?
11. What exactly does /device/remove-protection do at device and inverter level?
12. Is remove-protection reversible? Is it different from setting control to on?
13. What authorization level is required for remove-protection, and can it be disabled for BuiltRight's API key?

BuiltRight will not use the remove-protection endpoint until its precise effect, reversibility, authorization model, and safety implications are confirmed in writing.

5 Tamper, bypass, and connectivity alert requirements

BuiltRight previously discussed an immediate tamper-alert requirement with AshGridX. The intended behaviour is that unplugging a protected cable or disconnecting the AshGridX device from the inverter triggers an alert containing the identity of the affected customer device. BuiltRight will use that identity to locate the customer, site, order, and installation immediately.

The current documentation shows only the following example webhook:

```
{
  "type": "BYPASS",
  "deviceId": "string",
  "time": "string"
}
```

1. Does BYPASS mean physical tampering, cable disconnection, communication bypass, meter bypass, protection removal, or another condition?
2. Does unplugging the cable between the AshGridX device and inverter trigger BYPASS immediately?
3. Which protected cables, ports, sensors, or enclosure events can generate a tamper alert?
4. What is the maximum expected delay between physical tampering and webhook delivery?
5. Does loss of internet or device connectivity generate a separate DEVICE_OFFLINE or DISCONNECTED event?
6. Does reconnection generate TAMPER_RESTORED, DEVICE_ONLINE, or another restoration event?
7. Will the payload include deviceId, customerDeviceId, deviceNumber, siteId, and a human-readable tamper reason?
8. Can BuiltRight query the device immediately after an alert to verify live condition and connectivity?
9. How does AshGridX distinguish deliberate tampering from power failure, network outage, maintenance, installation, or equipment fault?
10. Is there a maintenance mode or alert-suppression window for approved technician work?
11. Are repeated or continuing tamper conditions re-notified or escalated?
12. Can historical tamper events be retrieved through an API endpoint?
13. Does the physical device retain and later transmit tamper events that occur while connectivity is unavailable?

Requested tamper event payload

```
{
  "eventId": "evt_12345",
  "type": "TAMPER_DETECTED",
  "reason": "INVERTER_CABLE_DISCONNECTED",
  "deviceId": "ashgrid-device-id",
  "customerDeviceId": "builtright-device-reference",
  "deviceNumber": "AGX-0001",
  "siteId": "site-reference",
  "occurredAt": "2026-08-11T10:30:00Z",
  "connectivity": "offline",
  "restored": false
}
```

6 Webhook verification, delivery, and security

The documentation specifies an x-ashgrid-signature header in the form `t={timestamp},v1={signature}` and describes HMAC SHA-256 verification. The precise signing algorithm and delivery guarantees are not documented.

1. What exact byte sequence is signed: `rawBody`, `timestamp + rawBody`, `timestamp + '.' + rawBody`, or another format?
2. Is the timestamp expressed in Unix seconds, Unix milliseconds, or ISO 8601 format?
3. What timestamp tolerance should BuiltRight apply to prevent replay attacks?
4. Must verification use the exact raw HTTP body before JSON parsing?
5. How are webhook secrets created, viewed, rotated, and revoked?
6. Can old and new secrets overlap safely during rotation?
7. Does every event contain a globally unique `eventId` for duplicate detection?
8. What HTTP status codes does AshGridX treat as successful webhook delivery?
9. How many times are failed webhooks retried, and what retry schedule is used?
10. Are webhook events delivered in order? Can duplicates or delayed events occur?
11. Can missed events be replayed from the developer dashboard or retrieved through an API?
12. What complete webhook event catalogue is supported, including control, connectivity, tamper, protection, and restoration events?
13. Can separate webhook URLs be configured for staging and production?
14. Can webhook sources be restricted by documented IP ranges in addition to signature verification?

7 API errors, limits, availability, and support

The documentation lists 200, 201, 204, 404, and 500 responses and refers to Java `ResponseStatusException` error objects. A production integration also requires documented authentication, validation, conflict, throttling, timeout, and service-availability behaviour.

1. What are the documented response bodies for validation errors and failed control operations?
2. How are 400, 401, 403, 409, 422, and 429 responses represented?
3. What request rate limits apply per API key, customer, or device?
4. Are rate-limit headers and `Retry-After` provided?
5. What client connection and response timeouts are recommended?
6. What retry policy is safe for GET requests and for device-control POST requests?
7. What uptime or service-level target applies to production control and webhook delivery?
8. Is there a public status page or incident-notification channel?
9. What logs or correlation identifiers should BuiltRight provide when opening a support case?
10. What are the support contacts and escalation hours for a live pilot or device-control incident?
11. Is data encrypted at rest, and what is the retention period for customer/device/control records?
12. Where is customer and device data hosted, and what data-protection terms apply?

Sensitive data observation

History responses include customer name, address, email, phone number, account number, owner identifiers, site identifiers, and device/vendor configuration. BuiltRight requests confirmation that API responses are scoped to its own company and that least-privilege access, retention controls, and audit logging are available.

8 Proposed BuiltRight pilot workflow

Step	Pilot test	Required result
1	Onboard and assign device	Identifiers map correctly to the BuiltRight customer, installation, and site.
2	Verify connectivity	Backend can identify online/offline condition and last-seen time.
3	Send control: off	Command is accepted and physical inverter state is independently confirmed.
4	Send control: on	Reactivation is confirmed and recorded without duplicate execution.
5	Disconnect protected cable	Immediate tamper event identifies the correct device, customer, and reason.
6	Restore cable	Restoration event arrives and incident state can be closed.
7	Simulate network outage	Offline event is distinguishable from tampering and queued events recover safely.
8	Exercise webhook failure	AshGridX retries and BuiltRight deduplicates/reconciles the event.
9	Review histories	Control, tamper, protection, and restoration audit trails are complete.

Operational safeguard

During the pilot, device disablement should require an authorized BuiltRight administrator after the customer's financing default and grace-period process has been verified and communications have been logged. Automatic disablement should only be considered after control confirmation, webhook reliability, identity mapping, replay protection, and exception handling have been proven.

9 Minimum acceptance criteria before production

- Device registration and customer/site assignment process documented.
- Stable mapping between deviceId and customerDeviceId confirmed.
- Live state, connectivity, and last-seen status available.
- Physical execution of on/off commands can be confirmed.
- Tamper and restoration events are explicitly defined and testable.
- Immediate cable-disconnection alert is demonstrated with the correct customer device identity.
- Webhook signature algorithm, raw-body handling, timestamp tolerance, and secret rotation are documented.
- Webhook event IDs, retry behaviour, replay/recovery method, and event catalogue are documented.
- Idempotency and offline-command behaviour are confirmed.
- Error responses, rate limits, timeouts, uptime expectations, and escalation contacts are provided.
- Remove-protection behaviour and authorization are explained and approved before use.
- Staging tests pass before production credentials or customer devices are enabled.

Requested response format

For each question or acceptance criterion, please indicate one of: Supported and documented, Supported but documentation pending, Not currently supported, or Planned with an expected delivery date. Links to updated documentation, example payloads, and test credentials may be supplied alongside the response.

BuiltRight is ready to proceed with a controlled pilot once the priority identity, control-confirmation, tamper-event, and webhook-security clarifications are resolved.

10 BuiltRight contact

Contact field	Details
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Thank you. We look forward to AshGridX's technical response and confirmation of the pilot setup process.